

# Sarvesh Gharat

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Incoming Visiting Researcher at MBZUAI (2026) — Multi-Agent Systems for Scientific Discovery

## EDUCATION

### IIT Bombay

*PhD in Artificial Intelligence and Data Science*

Mumbai, In

July 2022 – May 2027

### Vishwakarma Institute of Information Technology

*BTech in Electronics and Telecommunication*

Pune, In

Aug. 2018 – May 2022

## RESEARCH INTERESTS

My research focuses on online learning and sequential decision-making, with applications to reliable and resource-efficient LLM reasoning. My current interests also include continual learning and multi-agent systems for scientific discovery.

## EXPERIENCE

### • Summer Research Intern

May 2025 – July 2025

*Adobe Research*

*Host: Soumyabrata Pal, Ramasuri Narayanam*

- Improved LLM reasoning under fixed compute constraints by developing a novel method to inject auxiliary information during sequential test-time scaling.
- Designed and implemented an efficient, on-the-fly example selection strategy using embedding similarity, leading to more relevant in-context examples during inference.
- Focused on designing inference-time techniques that enhance model responses without changing any underlying parameters.

**Patent Filed: Gharat, S., Pal, S., Narayanam. R.,** Guided Sequential Test Time Scaling using additional hints from exemplars

### • Student Researcher

Oct 2024 – March 2025

*Google DeepMind*

*Host: Aparna Taneja, Milind Tambe*

- Evaluated AI-targeted intervention scheduling in a large-scale maternal health program using data from thousands of beneficiaries.
- Designed comparative analyses against matched control groups to measure effects on engagement, maternal health knowledge, and postnatal care behaviors.
- Developed robust analyses for noisy and variable survey responses, enabling reliable detection of intervention effects.

### • Visiting Researcher

Aug 2021 – June 2022

*Tata Institute of Fundamental Research*

*Host: Prof. Shriganesh Prabhu*

- Designed optimization methods to estimate the refractive index of optically thin materials using Terahertz Time Domain Spectroscopy.
- Implemented and benchmarked algorithms including Dual Annealing, SHGO, and Newton-Raphson to improve convergence stability.
- Built a user-friendly Python library to support experimental analysis and make the refractive index estimation pipeline accessible for both researchers and educators.

## PEER-REVIEWED PUBLICATIONS

- Singh, V., Ghosal, S., **Gharat, S.**, Pal, S., Narayanam. R., Manocha. D, ThinkRetrieve: Retrieval-Augmented Reasoning Traces for Test-Time Scaling, to appear in Findings of the Association for Computational Linguistics: EMNLP 2026
- Gharat, S.**, Karamchandani, N. and Nair, J., 2026, May. Cost-Aware Best Arm Identification via Dueling Feedback with Applications to Large Language Models. In Proc. of the 25th International Conference on Autonomous Agents and Multiagent Systems (pp. 3600-3602).

- Dasgupta, A.\*, **Gharat, S.\***, Madhiwalla, N., Hegde, A., Tambe, M., Taneja, A., 2025, October. AI-Targeted Calls Drive Measurable Improvements in Maternal Health, In Proc. of European Conference on Artificial Intelligence (pp. 5200-5207).
- **Gharat, S.**, Yadav, A., Karamchandani, N. and Nair, J., 2025, April. Representative arm identification: A fixed confidence approach to identify cluster representatives. In ICASSP 2025-2025 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP) (pp. 1-5). IEEE.
- Suresh, R., Bishnoi, H., Kuklin, A.V., Parikh, A., Molokeev, M., Harinarayanan, R., **Gharat, S.** and Hiba, P., 2024. Revolutionizing physics: a comprehensive survey of machine learning applications. *Frontiers in Physics*, 12, p.1322162.
- Bhatta, G., **Gharat, S.**, Borthakur, A. and Kumar, A., 2024. Gamma-ray blazar classification using machine learning with advanced weight initialization and self-supervised learning techniques. *Monthly Notices of the Royal Astronomical Society*, 528(1), pp.976-986.
- Mistry, P., Prasad, A., Maity, M., Pathak, K., **Gharat, S.**, Lekkas, G., Bhattarai, S., Kumar, D., Lissauer, J.J., Twicken, J.D. and Soubkiou, A., 2024. VaTEST III: Validation of eight potential super-earths from TESS data. *Publications of the Astronomical Society of Australia*, 41, p.e030.
- **Gharat, S.**, Borthakur, A. and Bhatta, G., 2024. Estimation of redshift and associated uncertainty of Fermi/LAT extragalactic sources with Deep Learning. *Monthly Notices of the Royal Astronomical Society*, 527(3), pp.6198-6210.
- **Gharat, S.**, Bose, B., Borthakur, A. and Mazumder, R., 2023. An image processing approach to identify solar plages observed at 393.37 nm by the Kodaikanal solar observatory. *RAS Techniques and Instruments*, 2(1), pp.393-397.
- Mistry, P., Pathak, K., Prasad, A., Lekkas, G., Bhattarai, S., **Gharat, S.**, Maity, M., Kumar, D., Collins, K.A., Schwarz, R.P. and Mann, C.R., 2023. VaTEST. II. Statistical Validation of 11 TESS-detected Exoplanets Orbiting K-type Stars. *The Astronomical Journal*, 166(1), p.9.
- Pininti, V.R., Bhatta, G., Paul, S., Kumar, A., Rajgor, A., Barnwal, R., and **Gharat, S.**, 2023. Exploring short-term optical variability of blazars using TESS. *Monthly Notices of the Royal Astronomical Society*, 518(1), pp.1459-1471.
- **Gharat, S.** and Dandawate, Y., 2022. Galaxy classification: a deep learning approach for classifying Sloan Digital Sky Survey images. *Monthly Notices of the Royal Astronomical Society*, 511(4), pp.5120-5124.
- Kulkarni, J.S., Cengiz, K. and **Gharat, S.**, 2021, March. Compact C-slot microstrip-fed planar antenna for wireless devices. In 2021 International Conference on Emerging Smart Computing and Informatics (ESCI) (pp. 651-654). IEEE.

\* *Equal contribution.*

## WORKSHOP PAPERS

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- **Gharat, S.**, Komiyama, J., 2026. PRISM: Problem Discovery via Structural Motifs in Knowledge Graphs. *ICML AI for Science Workshop 2026*.
- **Gharat, S.**, Borthakur, A. and Bhatta, G., 2023. Gamma Ray AGNs: Estimating Redshifts and Blazar Classification using Neural Networks with smart initialization and self-supervised learning. *NeurIPS Machine Learning and the Physical Sciences Workshop 2023*.

- **Gharat, S.**, Hebbalaguppe, R., Not All Thoughts Are Worth Thinking: Reliability-Aware Test-Time Scaling
- **Gharat, S.**, Komiyama, J., SGHA: Evidence-Grounded Research Problem Discovery with Local Language Models
- **Gharat, S.**, The Argmax Gap in Human Chess Move Prediction
- **Gharat, S.**, Chavan, U., Karamchandani, N. and Nair, J., Which Model to Train? A Budgeted Bandit Approach to Adaptive Fine-Tuning
- **Gharat, S.**, Karamchandani, N., and Nair, J., Cost-Aware Best LLM Identification using Dueling Feedback.
- Kumar, A., **Gharat, S.**, Beyond One-Shot Decisions: Progressive Morphological Reasoning in Galaxy Classification

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#### INVITED TALKS & PRESENTATIONS

- **Invited Talks**
  - Which Model to Train? A Budgeted Bandit Approach to Adaptive Fine-Tuning — Cohere for AI Reinforcement Learning Group (2026)
  - *Cost-Aware Best Arm Identification via Dueling Feedback with Applications to Large Language Models* — Cohere for AI Research Connections (2025)
  - *Representative arm identification: A fixed confidence approach to identify cluster representatives* — Cohere for AI Reinforcement Learning Group (2025)
  - *Estimating Redshifts of AGNs using Neural Networks* — IAU-IAA Astroinformatics Seminar (2024)
- **Poster Presentation:**
  - ACM India ARCS 2026
  - Google Research Week 2024
  - IndoML Graduate Forum 2024
  - ACM Pingala Interactions in Computing 2024

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#### ACADEMIC SERVICE & HIGHLIGHTS

- **Reviewer / Program Committee**
  - ICLR 2027
  - IJCAI AI for Social Good (2025, 2026) - Awarded Bronze Tier reviewer distinction in 2026
  - KDD AI for Sciences 2026
  - AAMAS 2025
  - IEEE Transactions on Information Theory
- **Research Grants and Support**
  - Awarded *Thinking Lab Research Grant* for the project on "An Online Learning Approach for Adaptive Fine-tuning"
  - Selected for the *Cohere Labs Catalyst Grant Program* for a proposal on online learning
  - Inaugural cohort grantee of the *Adaption Labs Research Grant*, for research on continual learning, online learning, and adaptive AI systems

- State Bank of India (SBI) funded project (2024–2026) on *Dynamic LLM Optimization and Fine-Tuning via Dueling Bandits*, exploring adaptive model selection under resource constraints

- **Selected Programs**

- AAMAS Doctoral Consortium 2026
- ACM India ARCS 2026
- Google Research Week (2024, 2025)
- IndoML Graduate Forum 2024
- ACM CODS-COMAD PhD Clinic 2024
- ACM Pingala Interactions in Computing 2024

- **Kaggle Expert**

- LLM Prompt Recovery Challenge: Silver Medal (Rank 79/2175)
- Santa 2025 Challenge: Bronze Medal (Rank 228/3357)

## TEACHING EXPERIENCE

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- **CS 747:** Foundations of Intelligent and Learning Agents (**Best TA Award**)
- **CS 663:** Digital Image Processing
- **EE 325:** Probability and Random Processes
- **DS 203:** Programming for Data Science

## MENTORSHIP EXPERIENCE

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- Algoverse AI Research Program 2026
- Krittika Astronomy Summer Project, IIT Bombay 2026
- Summer of Sciences, IIT Bombay 2023

## TECHNICAL SKILLS

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- **Programming:** Python, LaTeX, Apps Script
- **Libraries:** PyTorch, TensorFlow, NumPy, pandas, vLLM, Transformers
- **Tools:** Jupyter Notebook, Git, Docker, Google Cloud Platform, VS Code